Musical examples

The musical examples, including scores and audios, links to the official Analyzing Classical Form website.

Among the https://acf.research.mcgill.ca/supp_list.phpcomplete list of example we look at how the form inference model perform on only a few of them.

Model output

Identifying presentation

example 2-11

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We distinguish between bindings for idea and bindings for formal unit binding. For the presentation, it is the idea that is repeats, while for the period the repeating thing is the formal unit (ABAC)

```
Input surface:
  {a ⟦I None⟧} + {a ⟦IV None⟧}




Output analysis:
  size=12
  Presentation
    basicIdea= "a"
    occur1= UseIdeaWith   $0 ⟦I None⟧
    occur2= UseIdeaWith   $0 ⟦IV None⟧


  size=11
  {"a" ⟦I None⟧} + {"a" ⟦IV None⟧}
```

This example illustrates the basics of how variable binding leads to smaller size. By default, we assign a high cost of raw repetition markers from the input (as they are placeholders for the surface motivic ideas). On the other hand, calling a binding variable just cost 1.

example 2-12

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This time we make the input a little bit harder in that the basic idea is not an atomic marker, but the concatenation of two. And the model need to identify the repetition of “a”+“b” as the basic idea, and correctly build up the phrase parameter for the concatenated phrases.

```
 1 Input surface:
 2   {a ⟦I-V None⟧} + {b ⟦I-III None⟧} + {a ⟦VI-II None⟧} + {b ⟦VI-III None⟧}
 3
 4
 5
 6
 7 Output analysis:
 8   size=26
 9   Presentation
10     basicIdea= "a"+"b"
11     occur1= UseIdeaWith   $0 ⟦I-V None⟧+⟦I-III None⟧
12     occur2= UseIdeaWith   $0 ⟦VI-II None⟧+⟦VI-III None⟧
13
14
15
16   size=27
17   {"a" ⟦I-V None⟧}
18   +
19   {"b" ⟦I-III None⟧} + {"a" ⟦VI-II None⟧} + {"b" ⟦VI-III None⟧}
```

Identifying compound themes

example 6-1

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```
Input surface:
  {a ⟦I-V None⟧}
  +
  {b ⟦I-V-I-V Weak⟧}
  +
  {c ⟦IV-V-I None⟧}
  +
  {d ⟦VI-VII-I-V Weak⟧}
  +
  {a ⟦I-V None⟧} + {b ⟦I-V-I-V Weak⟧} + {c ⟦V None⟧} + {e ⟦I Strong⟧}



Output analysis:
  size=47
  Period
    antecedentHead= Continuation   {"a" ⟦I-V None⟧} {"b" ⟦I-V-I-V Weak⟧}
    antecedent= $0 + Continuation   {"c" ⟦IV-V-I None⟧} {"d" ⟦VI-VII-I-V Weak⟧}
    consequent= $0 + Continuation   {"c" ⟦V None⟧} {"e" ⟦I Strong⟧}



  size=56
  Period
    antecedentHead= {"a" ⟦I-V None⟧}
    antecedent= $0
                +
                {"b" ⟦I-V-I-V Weak⟧}
                +
                Continuation   {"c" ⟦IV-V-I None⟧} {"d" ⟦VI-VII-I-V Weak⟧}
    consequent= $0
                +
                {"b" ⟦I-V-I-V Weak⟧}
                +
                Continuation   {"c" ⟦V None⟧} {"e" ⟦I Strong⟧}
```

example 6-2

🖼️🔊 **Click to Play**

```
Input surface:
  {a ⟦I None⟧}
  +
  {b ⟦I None⟧}
  +
  {c ⟦I None⟧}
  +
  {a ⟦V None⟧}
  +
  {b ⟦V None⟧}
  +
  {c ⟦V None⟧}
  +
  {b ⟦I-VI None⟧} + {b ⟦IV-II None⟧} + {b ⟦V None⟧} + {d ⟦I-II-V-I Strong⟧}



Output analysis:
  size=62
  Continuation
    Presentation
      basicIdea= "a"+"b"+"c"
      occur1= UseIdeaWith   $0 ⟦I None⟧+⟦I None⟧+⟦I None⟧
      occur2= UseIdeaWith   $0 ⟦V None⟧+⟦V None⟧+⟦V None⟧
    Continuation
      Presentation
        basicIdea= "b"
        occur1= UseIdeaWith   $0 ⟦I-VI None⟧
        occur2= UseIdeaWith   $0 ⟦IV-II None⟧
      Continuation   {"b" ⟦V None⟧} {"d" ⟦I-II-V-I Strong⟧}



  size=63
  Sentence
    basicIdea= "a"+"b"+"c"
    presentation= Presentation
                    basicIdea= $0
                    occur1= UseIdeaWith   $0 ⟦I None⟧+⟦I None⟧+⟦I None⟧
                    occur2= UseIdeaWith   $0 ⟦V None⟧+⟦V None⟧+⟦V None⟧
    continuation= Continuation
                    Presentation
                      basicIdea= "b"
                      occur1= UseIdeaWith   $0 ⟦I-VI None⟧
                      occur2= UseIdeaWith   $0 ⟦IV-II None⟧
                    Continuation   {"b" ⟦V None⟧} {"d" ⟦I-II-V-I Strong⟧}
```

example 6-5

🖼️🔊 **Click to Play**

```
Input surface:
  {a ⟦I-V None⟧}
  +
  {a ⟦V-I None⟧}
  +
  {b ⟦IV-I None⟧}
  +
  {b ⟦IV-I None⟧} + {c ⟦II None⟧} + {c ⟦I None⟧} + {d ⟦V Weak⟧}



Output analysis:
  size=46
  Continuation
    Presentation
      basicIdea= "a"
      occur1= UseIdeaWith   $0 ⟦I-V None⟧
      occur2= UseIdeaWith   $0 ⟦V-I None⟧
    Continuation
      {"b" ⟦IV-I None⟧}
      Continuation
        {"b" ⟦IV-I None⟧}
        Continuation
          {"c" ⟦II None⟧} Continuation   {"c" ⟦I None⟧} {"d" ⟦V Weak⟧}



  size=47
  Sentence
    basicIdea= "a"
    presentation= Presentation
                    basicIdea= $0
                    occur1= UseIdeaWith   $0 ⟦I-V None⟧
                    occur2= UseIdeaWith   $0 ⟦V-I None⟧
    continuation= Continuation
                    {"b" ⟦IV-I None⟧}
                    Continuation
                      {"b" ⟦IV-I None⟧}
                      Continuation
                        {"c" ⟦II None⟧}
                        Continuation   {"c" ⟦I None⟧} {"d" ⟦V Weak⟧}
```

Others

supp 6-1

🖼️🔊 **Click to Play**

example 5-11

🖼️🔊 **Click to Play**